

Lab activity: Capturing web traffic in Coursera's Linux desktop

Learning objective

Learn how to open Wireshark, start a capture, browse a website and analyse the traffic.

Part 1: Open Wireshark

- 1: On the Linux Desktop, find Wireshark in the Applications menu (blue shark fin icon).
2. Double-click to open it.
3. You will see the "Welcome to Wireshark" screen with a list of network interfaces.

Part 2: Choose the Correct Network Interface

1. Look for the interface that is active, it usually shows a graph moving up and down.
 - Often it is called eth0 or similar.
 - Do not select lo (loopback) for this activity.
2. Double-click that interface to start capturing. You should now see packets scrolling in real-time.

Part 3: Generate Web Traffic

1. Without closing Wireshark, open Google Chrome in the Linux Desktop.
2. **Important step:** A default Chrome page should appear. **From there, open a new Browser tab.** That will allow you to navigate to various websites.
3. Go to one of the websites **by typing the link directly in the URL bar (not via Google).** For example, <https://www.coursera.org> or any site the lab allows.
4. Spend about 30 seconds browsing:
 - Click a link or two.
 - Refresh the page once.

Part 4: Stop the Capture

1. Go back to Wireshark.
2. At the top, click the red square button (Stop).
3. You now have a saved list of packets from your browsing session.

Part 5: Apply Display Filters

1. At the top of Wireshark, find the Display Filter box (it says “Apply a display filter...”).
2. Type the following filters one at a time, press Enter, and observe the results:
 - http → Shows HTTP requests (if the site uses plain HTTP).
 - tls → Shows encrypted traffic (most modern websites use HTTPS, which appears as TLS).
 - dns → Shows the DNS queries your browser made to look up website names.

Part 6: Explore a Packet

1. Pick one packet in the middle pane.
2. Click the arrow icons to expand the layers:
 - Ethernet II → shows MAC addresses.
 - Internet Protocol (IP) → shows source and destination IP addresses.
 - Transmission Control Protocol (TCP) → shows ports and flags (SYN, ACK, etc.).
 - Hypertext Transfer Protocol (HTTP) → if available, shows the request (e.g., GET /).
3. Ask yourself:
 - Who is the Source IP? (likely your VM).
 - Who is the Destination IP? (the web server).
 - What protocol is used?

Part 7: Follow a TCP Stream

1. Right-click any packet where the protocol is HTTP or TLS.
2. Select “Follow → TCP Stream.”
3. A window opens showing the conversation between your computer and the server.
 - If it’s HTTP: you will see readable requests and responses.
 - If it’s HTTPS/TLS: you will see encrypted data, but still the communication flow.
4. Close the window when finished.

Part 8: Save Your Work

1. Go to File → Save As....
2. Save your capture as web_traffic_lab.pcap. If you want this file to persist for future reference (next time you open the lab), please move the file inside the **exercise** folder that you see on the Desktop.
3. You can open this file again later for practice (only if you saved it inside the **exercise** folder on the Desktop).

Reflection Questions

1. What is the difference between HTTP traffic and TLS (HTTPS) traffic in Wireshark?
2. What is the source IP address of your VM and what is the destination IP address of the website?
3. Did you find any DNS queries? What website name was resolved?
4. When you followed a TCP stream, what did you notice? Was it readable or encrypted?